

SANJAY BOLLINI

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EDUCATION

University of Washington

B.S. Computer Science | Minor: Applied Mathematics | GPA: 3.75 | Dean's List

June 2028

Seattle, WA

Relevant Coursework: CSE 351: Computer Systems • CSE 311: Foundations of Computing • AMATH 352: Numerical Linear Algebra • CSE 332: Data Structures & Parallelism (upcoming) • CSE 333: Systems Programming (upcoming)

Certifications: Building with the Claude API — Anthropic • Multi-Agent Systems — Google • AWS Cloud Essentials • AI/Neuroscience — UW

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, SQL, HTML/CSS

AI / ML: LLMs (Claude, OpenAI), Spiking Neural Networks, CNNs, PyTorch, Prompt Engineering, Claude Code, GitHub Copilot

Frameworks & Tools: Next.js, Vue.js, Node.js, PostgreSQL, Git, AWS, x86-64, Static Analysis (JayHorn, JBMC, CBMC)

EXPERIENCE

Alliance Endocrinology

Feb 2020 – Present

IT Manager

Southlake, TX

- Sole technical owner of clinic IT for 5+ years: networks, hardware, EHR systems, security infrastructure, and vendor relationships — independently resolving complex issues under production constraints.
- Deployed Sophos firewall, IP cameras, and endpoint security software; enforced HIPAA compliance across all systems.
- Designed and maintained clinic website in HTML/CSS; implemented SEO and Google Ads strategy to improve patient acquisition.

Mouri Tech

Jul 2024 – May 2025

Data Engineering Intern

Irving, TX

- Built SQL analytics pipelines with structured debugging and SDLC practices across planning, testing, and deployment phases.

RESEARCH

UT Dallas — Software Testing & Analysis Lab

Jun – Aug 2024

Static Analysis Research Intern

Dallas, TX

- Integrated ML into static analysis tools (JayHorn, JBMC, CBMC) to automate Java program verification across benchmark suites (Fibonacci, McCarthy 91, Ackermann).
- Designed test pipelines with independent variance analysis; presented findings at a lab research showcase.

UW Machine Learning Research Club

Sep 2025 – Present

Spiking Neural Networks Researcher

Seattle, WA

- Collaborate on bio-inspired SNN architectures targeting energy-efficient inference; applying neuromorphic computing principles to real-world ML problems.

PROJECTS

BillPilot AI

2026

Next.js 15, TypeScript, Claude API, PostgreSQL

- Full-stack AI medical billing assistant with multi-step agentic pipeline: document ingestion → LLM extraction → structured output → user review loop. Submitted to OpenAI × Handshake Codex Creator Challenge.
- Includes Zod validation, Vitest unit tests, and Playwright E2E tests — demonstrating end-to-end software engineering rigor.

AI Market Intelligence System

2026

Python, PostgreSQL, Claude Haiku/Sonnet, yfinance, cron | github.com/Zing302/ai-market-intel

- 4-agent pipeline (Analyst, Researcher, Trader, Manager) runs 3x/day on 20 tracked symbols — producing BUY/HOLD/SELL signals with conviction scores, SMA/RSI technicals, and Claude sentiment analysis.
- Ingests SEC 8-K filings, 800+ daily news headlines, and real-time prices; delivers a structured daily briefing email at 5pm ET. 141 unit tests, fully automated via cron.

Quantum Password Generator

2025

Python, FastAPI, Qiskit, IBM Qiskit Aer, HTML/CSS | github.com/Zing302/quantum-password-generator

- Applies Hadamard gates across multiple qubits to generate true-random entropy via measurement collapse — replacing PRNG with quantum randomness. Deployable on real IBM quantum hardware.
- Built a FastAPI backend serving quantum-generated passwords to an HTML/CSS frontend; full-stack interface for real-time password generation and strength analysis.

Mindfulness4Success

2023

Vue.js, JavaScript | mindfulness4success.com

- Solo-built wellness web app with guided meditation, habit tracking, and community resources
- Presented at Youth Entrepreneur Summit
- Donated proceeds to Carroll ISD CS Honor Society.